Dimensional Shift: Pulse-Driven Reservoir-Wavelet Architecture for Ultra-Low-Power Adaptive Signal Processing

Stylized technical working document for further development into a bachelor's / master's thesis track

Author

Bc. Jiří Burda

Faculty of Applied Informatics, Tomas Bata University in Zlín (FAI UTB)

Date prepared: 17 April 2026

Note: The document is formatted as a representative working version with a formal structure, language editing, and a reference list prepared in a style close to ČSN ISO 690.

Abstract

This work presents the Dimensional Shift (DS) architecture, a pulse-driven and adaptive system for processing streaming signals on energy-constrained platforms. The proposed approach combines the principles of reservoir computing, wavelet packet decomposition, dynamic pruning, and streaming Top-K selection. The central idea is to explicitly manage dimensionality as a constrained resource through an Expand-Collapse cycle, in which the expanded phase creates a rich time-frequency representation and the collapse phase actively reduces the number of useful dimensions. The architecture targets scenarios in which low energy consumption, online adaptation, and fast response must be combined, such as edge AI, biomedicine, predictive maintenance, or neuromorphic systems. Initial experiments on a real 0-230 V voltage signal indicate promising potential for adaptive signal processing in a low-power edge inference regime.

Keywords: Dimensional Shift, reservoir computing, echo state network, liquid state machine, wavelet packet decomposition, RigL, Space-Saving, edge AI, neuromorphic hardware.

Main Contributions of the Proposal

- explicit dimensionality control via a pulse-driven Expand-Collapse cycle
- combination of reservoir computing and wavelet packet decomposition for local and temporal representation
- streaming dimensionality reduction using Space-Saving Top-K and RigL-type dynamic sparsity
- orientation toward ultra-low-power edge AI and natural mapping to neuromorphic hardware

1. Introduction

Current edge AI development emphasizes the ability to perform inference as close as possible to the point where data are generated, with low latency, limited memory, and a very strict energy budget [6]. Traditional deep models such as LSTM, CNN, and transformers achieve very high accuracy in suitable tasks, but at the cost of substantial demands on compute, memory, and data movement. In many real deployments - such as industrial sensors, wearable devices, or autonomous systems - there is therefore a need for architectures that are adaptive while remaining operationally efficient.

Dimensional Shift is designed as an alternative to fully trained deep networks. Its center of gravity is not massive gradient-based training of the whole network, but rather the rich dynamics of a reservoir, local time-frequency analysis, and continuous dimensionality control. The goal is not only classification or prediction, but also the ability to work stably with streaming signals, adapt to changes, and suppress unnecessary computation.

2. Theoretical Background

Reservoir computing was formally established at the beginning of the millennium in two classical lines: Herbert Jaeger's Echo State Networks [1] and Wolfgang Maass, Thomas Natschläger, and Henry Markram's Liquid State Machines [2]. In both cases, the idea is that the input signal is projected into a high-dimensional

dynamical space, while typically only the output layer is trained. This radically reduces training complexity while preserving the ability to represent complex temporal dependencies.

For processing nonlinear and nontrivially non-stationary signals, multi-scale analysis is also important. Wavelet packet decomposition makes it possible to decompose a signal into locally defined time-frequency components and to select suitable bases according to an information criterion [3]. This property is especially valuable where the classical FFT does not provide sufficient sensitivity to short transients or local spectral changes.

The third building block of the proposal is the use of sparsity and streaming selection. The Space-Saving algorithm provides an efficient way to approximate the most important elements in data streams and is suitable for online Top-K selection [4]. RigL represents a modern approach to dynamic sparsity, in which the topology of connections changes during training without the need to materialize a fully dense network [5]. The combination of these principles creates the methodological basis for active dimensionality control in DS.

3. Comparison with Classical Methods

From the perspective of signal processing, DS can be understood as a synthesis of the advantages of several approaches. FFT is efficient and interpretable, but it operates globally and is less effective at capturing local anomalies. Wavelet methods provide time-frequency localization, but by themselves they do not address adaptive pulse-driven dynamics. Deep neural models are highly expressive, but in edge environments they often run into limits of power consumption, memory, and latency. Reservoir computing, in contrast, offers rich dynamics with relatively cheap readout training [1-2, 11].

Dimensional Shift builds on this foundation by treating dimensionality not as a fixed property of the model, but as an actively managed resource. The Expand phase increases the representation and improves the separability of signal states; the Collapse phase then retains only the informative parts of the representation. In this way, DS differs from both purely analytical methods and classical DNN architectures.

Comparison of Basic Approaches

Approach	Strengths	Limits	Contribution to DS
FFT	fast global spectral analysis	weaker sensitivity to transients	baseline frequency-domain view
Wavelet packet	local time-frequency decomposition	no intrinsic adaptive dynamics	improves sensitivity to local changes
DNN (LSTM/CNN/Transformer)	high expressiveness and accuracy	higher power, memory, and latency demands	reference performance level
Reservoir computing	cheaper readout training, rich dynamics	sensitivity to reservoir configuration	core dynamical component of DS
Dimensional Shift	adaptive dimensionality and pulse-driven behavior	detailed pipeline validation required	combines the strengths of the above approaches

Table 1: Qualitative comparison of methods relevant to the DS proposal.

4. Dimensional Shift Architecture

The core of the proposal is an explicit Expand-Collapse cycle controlled by the Te/Tc pulse. During the Expand phase, the input stream is processed by a combination of reservoir computing and wavelet packet decomposition, producing a rich state space that captures both local and global signal dynamics. The subsequent Collapse phase applies streaming Top-K selection and dynamic pruning so that only information-bearing dimensions remain.

The architecture also assumes a higher-level control layer called the meta-colony and a pain reflex mechanism. Their role is to adaptively change system parameters when signal 'pain' is detected, i.e. a state in which the current representation is no longer sufficient for stable inference. The Dimension Bus then serves as a replayable communication protocol controlled by a configuration identifier (cfg_id), enabling repeatability, auditability, and management of experimental variants.

This proposal is interesting because it combines three levels of adaptivity: representational adaptivity, topological adaptivity, and control-parameter adaptivity. This multi-layered adaptivity gives DS the potential to react to signal drift, changes in operating regime, and sudden anomalies without having to launch a full retraining of the entire architecture.

Data-Flow and Adaptation Logic

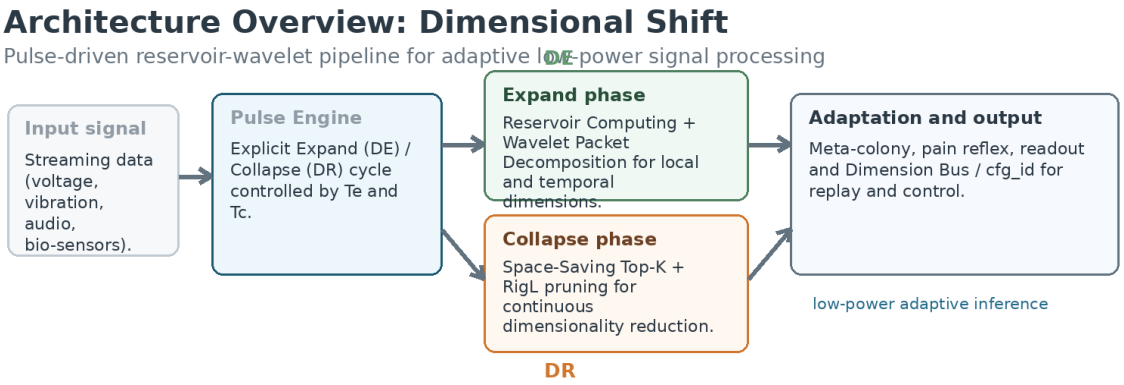


Figure 1: Simplified architecture diagram of Dimensional Shift.

Layer	Function	Expected effect
Pulse Engine	controls DE ↔ DR switching	controlled expansion and reduction of the representation
Reservoir + wavelets	generate a rich state space	higher sensitivity to local and temporal patterns
Space-Saving + RigL	retain only relevant dimensions	lower computational and memory load
Meta-colony / pain reflex	tune parameters during regime changes	greater robustness to drift and anomalies
Dimension Bus	transfers configurations and state replay	auditability and reproducibility

Table 2: Overview of the main layers and functions of the DS architecture.

5. Energy Efficiency and Relevance to Edge AI

In edge scenarios, the limiting factor is often not maximum accuracy at any cost, but a compromise between accuracy, latency, battery life, and computational locality [6]. Dimensional Shift is designed so that most computation takes place in fixed or only limitedly adapted structures, while more intensive decision-making is concentrated in a lighter output layer and dimensionality-control mechanisms. This philosophy is aligned with the main motivation of reservoir computing as well as the development of neuromorphic systems [7-11].

The expected advantage of the architecture is that, with suitable mapping, it can significantly reduce the number of unnecessary operations, transfers, and active parameters. Compared with dense DNNs, this is not only a reduction in the number of MAC operations, but also a shift in the computational paradigm toward sparse, event-driven, and continuously adaptive inference. This is precisely the type of space in which platforms such as Loihi, SpiNNaker, and Akida have been gaining long-term relevance [7-9, 12].

6. Preliminary Experimental Results

The experimental part is based on a real voltage signal in the range 0-230 V, stored in the file voltage_default.csv. According to the supplied summary, variants based on random kernels, wavelet packet decomposition, a reservoir approach, and hybrid strategies were compared. Evaluation in the arena used the metrics RMSE, objective, and estimated energy cost.

Because the detailed numerical leaderboard was not included in the supplied materials, this version of the document formulates only qualitative conclusions. First, hybrid strategies that combine a dynamic state representation with local time-frequency analysis appear more promising than isolated baseline methods. Second, active dimensionality reduction has the potential to improve operational efficiency without losing adaptive capability. Third, working with an explicit pulse appears to be a suitable mechanism for stabilizing streaming-data processing. In the next version of the text, it is advisable to add the full experimental table, reproducible settings, and error-curve plots.

Recommended Metrics for the Next Version of Experiments

Metric	Meaning	Note for interpretation
RMSE	reconstruction / prediction error	report together with variance across runs
Objective	aggregate optimization function	separate accuracy from the energy component
Energy estimate	operational suitability for edge	tie to a specific hardware or MCU profile
Adaptation latency	reaction speed to drift or anomaly	monitor during changes in signal regime
Sparsity ratio	degree of reduction in active dimensions	link to accuracy and stability

Table 3: Metrics recommended for systematic evaluation in the next version of experiments.

7. Potential Application Areas

The DS architecture has clear application potential in tasks where data arrive continuously, the signal has variable dynamics, and computational requirements must be kept to a minimum. These include event-based vision and processing data from DVS cameras, predictive maintenance of machines, wearable biomedicine (for example ECG or EEG), audio processing at the edge, smart agriculture, structural health monitoring, and autonomous robotic platforms.

The common denominator of these domains is that the system does not have the luxury of constant cloud support or room for heavy retraining. An adaptive, sparse, and locally deployable architecture therefore represents a relevant research direction as well as a practical engineering opportunity.

8. Mapping to Neuromorphic Hardware

Dimensional Shift is conceptually compatible with neuromorphic hardware. On Intel Loihi, event-driven communication, local plasticity, and the open Lava software stack can be used for experimental mapping and rapid prototyping [7-10]. SpiNNaker is relevant mainly because of its emphasis on massively parallel simulation of spiking networks and low-power operation [9]. In the case of BrainChip Akida, the most interesting aspects are its orientation toward event-based processing, sparsity, and edge deployment [12].

This is not merely a matter of technical compatibility, but of computational philosophy. DS does not assume a permanently active dense network; instead, it works with a continuously managed flow of representations and selective activation of useful dimensions. Such a regime is naturally closer to neuromorphic architectures than to classical accelerators optimized for dense matrix multiplication.

9. Conclusion and Future Work

Dimensional Shift represents a promising direction for ultra-low-power adaptive signal processing in edge environments. Its contribution does not lie in a single new component, but in a considered combination of reservoir computing, wavelet-based multi-scale analysis, streaming selection, and dynamically controlled sparsity. This yields an architecture with the potential to combine adaptivity, interpretability, operational efficiency, and good mappability to neuromorphic hardware.

From the perspective of future work, three immediate steps are suggested. The first is standardization of the experimental methodology, including publicly described datasets, baseline models, and metrics. The second is the implementation of a fully reproducible pipeline, including visualization of internal states and an auditable Dimension Bus protocol. The third is systematic verification of mapping to neuromorphic platforms and development of a high-value application scenario, for example in industrial diagnostics or biomedicine. In both academic and technology-transfer contexts, DS can therefore form a solid basis for a follow-up master's thesis and potentially also for a technology spin-off.

References

[1] JAEGER, Herbert. The “echo state” approach to analysing and training recurrent neural networks. Bonn: German National Research Center for Information Technology, 2001. GMD Report 148.

[2] MAASS, Wolfgang; NATSCHLÄGER, Thomas and MARKRAM, Henry. Real-Time Computing Without Stable States: A New Framework for Neural Computation Based on Perturbations. Neural Computation. 2002, 14(11), 2531-2560. DOI 10.1162/089976602760407955.

[3] COIFMAN, Ronald R. and WICKERHAUSER, Mladen V. Entropy-based algorithms for best basis selection. IEEE Transactions on Information Theory. 1992, 38(2), 713-718. DOI 10.1109/18.119732.

[4] METWALLY, Ahmed; AGRAWAL, Divyakant and EL ABBADI, Amr. Efficient Computation of Frequent and Top-k Elements in Data Streams. In: EITER, Thomas and LIBKIN, Leonid, eds. Database Theory - ICDT 2005. Berlin: Springer, 2005, pp. 398-412. Lecture Notes in Computer Science, vol. 3363. DOI 10.1007/978-3-540-30570-5_27.

[5] EVCI, Utku et al. Rigging the Lottery: Making All Tickets Winners. In: Proceedings of the 37th International Conference on Machine Learning. Proceedings of Machine Learning Research. 2020, 119, 2943-2952.

[6] SINGH, Raghubir and GILL, Sukhpal Singh. Edge AI: A survey. Internet of Things and Cyber-Physical Systems. 2023, 3, 71-92. DOI 10.1016/j.iotcps.2023.02.004.

[7] DAVIES, Mike et al. Advancing Neuromorphic Computing With Loihi: A Survey of Results and Outlook. Proceedings of the IEEE. 2021, 109(5), 911-934. DOI 10.1109/JPROC.2021.3067593.

[8] INTEL LABS. Taking Neuromorphic Computing to the Next Level with Loihi 2 Technology Brief [online]. 2021 [accessed 2026-04-17]. Available from: https://download.intel.com/newsroom/2021/new-technologies/neuromorphic-computing-loihi-2-brief.pdf

[9] PAINKRAS, Eustace et al. SpiNNaker: A 1-W 18-core system-on-chip for massively-parallel neural network simulation. IEEE Journal of Solid-State Circuits. 2013, 48(8), 1943-1953. DOI 10.1109/JSSC.2013.2259038.

[10] LAVA. Lava Software Framework [online]. 2026 [accessed 2026-04-17]. Available from: https://lava-nc.org/

[11] YAN, Min et al. Emerging opportunities and challenges for the future of reservoir computing. Nature Communications. 2024, 15, article 2056. DOI 10.1038/s41467-024-45187-1.

[12] POSEY, Brien M. What Is the Akida Event Domain Neural Processor? [online]. BrainChip, 2020 [accessed 2026-04-17]. Available from: https://brainchip.com/wp-content/uploads/2020/03/BrainChip_tech-brief_What-is-Akida_v3-1.pdf

Citation note: The reference list is formatted in a style close to ČSN ISO 690. Before submitting the final academic version, it is advisable to align the template, punctuation, and any departmental rules for DOI and online sources.